

QUESTION BANK

MODULE 2 – QUANTUM MECHANICS

1. What are Matter waves? Mention the properties of Matter waves.
2. State de Broglie hypothesis. Show that the de Broglie wavelength of an electron is found to be $\frac{12.26}{\sqrt{V}} \text{ \AA}$.
3. State and explain Heisenberg's uncertainty principle also discuss its physical significance.
4. Using uncertainty principle, prove that free electron does not exist inside the nucleus.
5. Explain Principle of Complementarity.
6. What is wave function? Explain the physical significance (Probability density and Normalization) of wave function.
7. Set up one dimensional time-independent Schrodinger's wave equation.
8. Discuss the Expectation value, Eigen values and Eigen functions.
9. Derive an expression for energy Eigen value and Eigen function for a particle in a potential well of infinite depth hence obtain the Normalised Eigen function.
10. Discuss the Eigen function, Eigen values and probability density with energies for a particle in an infinite potential well for ground and first two excited states.